

Project Gutenberg language diversity over time

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Abstract

context for understanding the evolution of digital preservation efforts.

1 Introduction

This study analyses the linguistic diversity in Project Gutenberg's digital library collection over time, examining changes in both English and non-English books added to the Gutenberg Collection per year. It aims to answer what the difference is in the proportion of languages added to Project Gutenberg over time. This research analyses the works added year over year to the Gutenberg collection, showing the changes in addition rates of differing languages. The hypothesis is that over time, the proportion of English books will decrease compared to non-English books, reflecting an increase in internet availability and interest in preservation of historical books, and a decrease in available public domain books written in English.

2 Related Work

Several studies have examined linguistic diversity in digital libraries and Project Gutenberg specifically:

1. **whitt`diachronic`2018** explored multilingualism on the web and Project Gutenberg's early efforts at linguistic diversification, providing important historical

2. **gerlach`standardized`2020** analysed a curated Project Gutenberg database, revealing a strong bias toward Western languages, with English comprising 81% of all books. Their work established baseline metrics for understanding linguistic representation in digital libraries.

Project Gutenberg also knows about the disparity in languages in its collection. For example, on the frequently asked questions pages on the [gutenberg.org](https://www.gutenberg.org) website they explain how books are added, namely by volunteers. This makes it difficult for Project Gutenberg to directly change which books are added, as they can't dictate what these volunteers add.

3 Data

Pre-processing The data pre-processing was done by downloading the metadata of all Project Gutenberg books from the [gutenberg.org](https://www.gutenberg.org) website on 06-01-2025. This data can be downloaded in the form of a CSV file, which contains the relevant fields for this article: the date added and the language. Books that included multiple languages, either due to being a translation that includes the original text or being a translation dic-

Language	1970-1980	1881-1990	1991-2000	2001-2010	2011-2020	2021-2024
English	9	1	1861	27375	22767	7970
Other	0	0	79	5526	6703	2759

Table 1: Books added to Project Gutenberg per decade

tionary, are counted for both languages. The python script used for extracting the data will be included with the paper. Due to not being completed, and thus giving incomplete information, the year 2025 was excluded. As only the language codes were given with the metadata, language codes were manually linked to their respective language. This resulted in a matrix for book additions per year for every language, from the years 1971 to 2024, with 70 languages. The the non-English books were all summed, creating the distribution graph shown in figure 1. Books added per decade are shown in table 1. The complete data matrix is shown in table 2 in the Appendix, with the total books per language shown in figure 2, also in the appendix.

4 Results

As seen in figure 1, the proportion of English of non-English books does change, especially after the peak and consequent collapse of English books around 2004, with an unsteady decline after that point. non-English books

Discussion

5 Conclusion

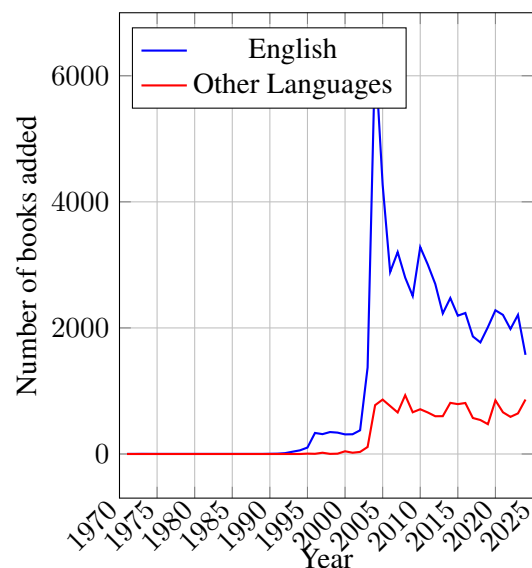


Figure 1: Number of books added to Project Gutenberg per year for English and non-English languages

6 Appendix

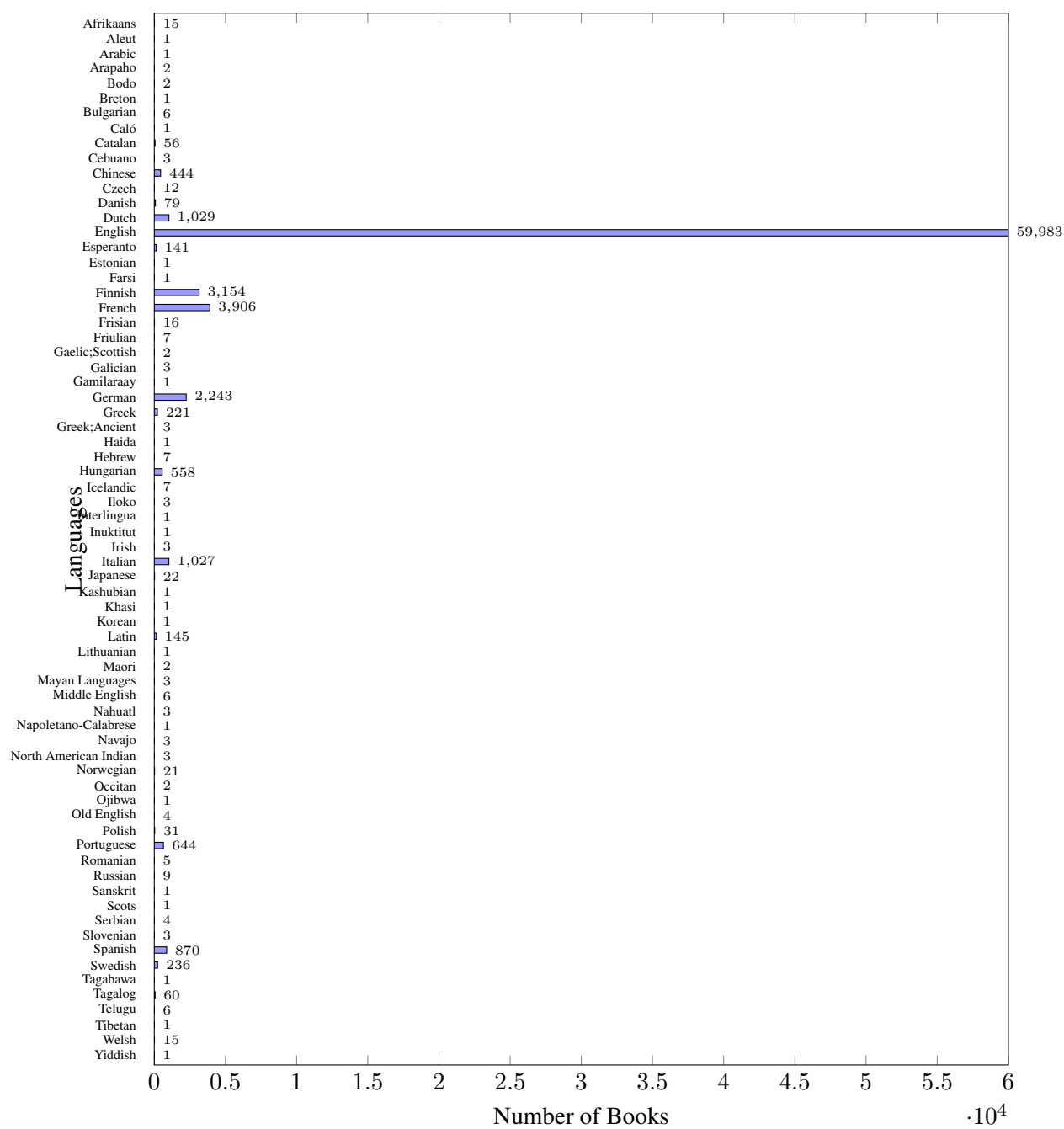


Figure 2: Total books per language in the Gutenberg Corpus

Language	1971	1972	1973	1975	1976	1977	1978	1979	1980	1991	1992	1993	1994	1995	1996	1997	1998	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023	2024	
English	1	1	2	0	0	0	1	1	1	5	13	36	58	102	334	315	348	339	311	313	376	1373	6363	4273	2887	3203	2798	2507	3382	3007	2230	2474	2194	2237	1865	1770	2013	2280	2007	1983	2207	1573		
French	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	2	1	6	1	39	328	263	319	163	213	187	185	239	188	191	185	112	129	79	110	84	200	160	133	208	207		
German	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	114	142	91	65	64	57	41	82	61	15	286	257	326	230	154	150	210	129	135	229	305			
Italian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	26	116	139	91	65	64	57	41	82	61	15	286	257	326	230	154	150	210	129	135	229	305		
Dutch	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	26	116	139	91	65	64	57	41	82	61	15	286	257	326	230	154	150	210	129	135	229	305		
Spanish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	80	71	87	62	70	76	54	55	40	36	40	25	37	25	82	35	65	42	30	36	45			
Portuguese	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	8	15	12	61	24	44	52	40	55	39	100	116	87	35	10	41	58	60	37	24	56	45		
Hungarian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	7	55	49	27	24	52	52	29	21	29	33	43	52	57	45	41	33	67	37	48	27	28		
Chinese	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	6	1	2	7	6	0	46	32	10	1	5	4	3	3	35	1	32	17	5			
Swedish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	10	3	55	0	78	247	11	0	0	2	3	0	6	38	31	23	19	12	9	11	1	6	7
Greek	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	9	14	9	5	11	7	0	5	0	6	38	31	23	19	12	9	11	1	6	7			
Hebrew	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	3	1	17	9	11	10	5	7	4	5	5	12	7	2	7	5	2	2	2	5	7	2		
Japanese	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	2	4	16	16	17	7	9	10	3	7	8	9	3	2	1	1	0	1	4	4	1	2	4
Norwegian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	13	1	10	3	2	12	8	10	9	3	1	1	0	1	0	1	0	1	0	0	0	0	
Finnish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Danish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Tagalog	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Catalan	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Polish	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Russian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Ukrainian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Belarusian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Georgian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Armenian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Ossetian	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	
Abkhaz	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0</																								

Table 2: The number of books added to Project Gutenberg per year per language